

The Geography of Dining: Neighborhood Income and Restaurants in Chicago

Jessica Xu



Project Description

Map Chicago's "geography of dining" by linking restaurant features to neighborhood socioeconomic status (SES)

- SES measure: median household income (community area)
- Test whether higher-SES neighborhoods differ in
 - restaurant prices (price level)
 - broader characteristics such as restaurant type, service options (takeout, delivery, dine-in, reservable), opening-hours, ratings, and review volume

RQ1: How are restaurant prices (price level) associated with neighborhood socioeconomic status (SES), measured by median household income, across Chicago?

- H1: Higher-SES neighborhoods → higher average price levels + a greater concentration of expensive restaurants

RQ2: Do restaurant characteristics (type, service options, hours, ratings, and review count) vary systematically across neighborhoods with different socioeconomic status?

- H2: Higher-SES neighborhoods will have:
 - More full-service and reservable restaurants
 - Greater diversity of restaurant types
 - Higher average ratings and review counts

Why it matters:

- Evidence of spatial inequality
- Reflects neighborhood stratification and commercial development
- Reveals how consumption patterns shape urban space

Data

Source	Sample Size	Time Period	Collection Method	Key Variables Created
Google Maps	3,388 → 1,968 matched restaurants	2026	Google Places API	price_level, rating, review_count, service flags (takeout, delivery, dine-in, reservable), primary_type
Yelp	5,112 → 1,968 matched restaurants	2026	Yelp Fusion API	price_level, rating, review_count, service flags (takeout, delivery, dine-in, reservable), primary_type
ACS 5-Year Data (Chicago Data Portal)	77 community areas	2018–2022	Download	median_household_income
Chicago Community Area Boundaries (Chicago Data Portal)	77 polygons	current shapefile	Download	spatial join to assign restaurants to community areas

Data Quality & Validation

- **Coverage:**
 - All 77 community areas have restaurant observations (no spatial gaps)
- **Deduplication:**
 - Deduplicated by place_id (Google) and business_id (Yelp)
 - kept only operational restaurants
- **Boundary filtering:**
 - Google filtered by Chicago boundary polygon
 - Yelp filtered by Chicago ZIP codes + IL state
- **Cross-platform matching:**
 - BallTree + Jaro-Winkler similarity
 - 1,968 matched pairs (score $\geq$ 80)
 - lowest-scoring matches verified by address
- **Missingness:**
 - merging across platforms reduces gaps
 - ~8% missing price level, ~3% missing rating for merged dataset
- **Cross-validation:**
 - Two independent data sources with matching names/locations increases confidence in data quality

Data Pipeline: Google API → Yelp API → cleaning → matching → ACS merge → analysis

Lessons Learned and Improvements

Challenge:

- Limited API calls forced tradeoffs between coverage and depth
- Nearby Search returns a maximum of 20 results, which can bias spatial coverage



scraping around one point



ZIP 60637



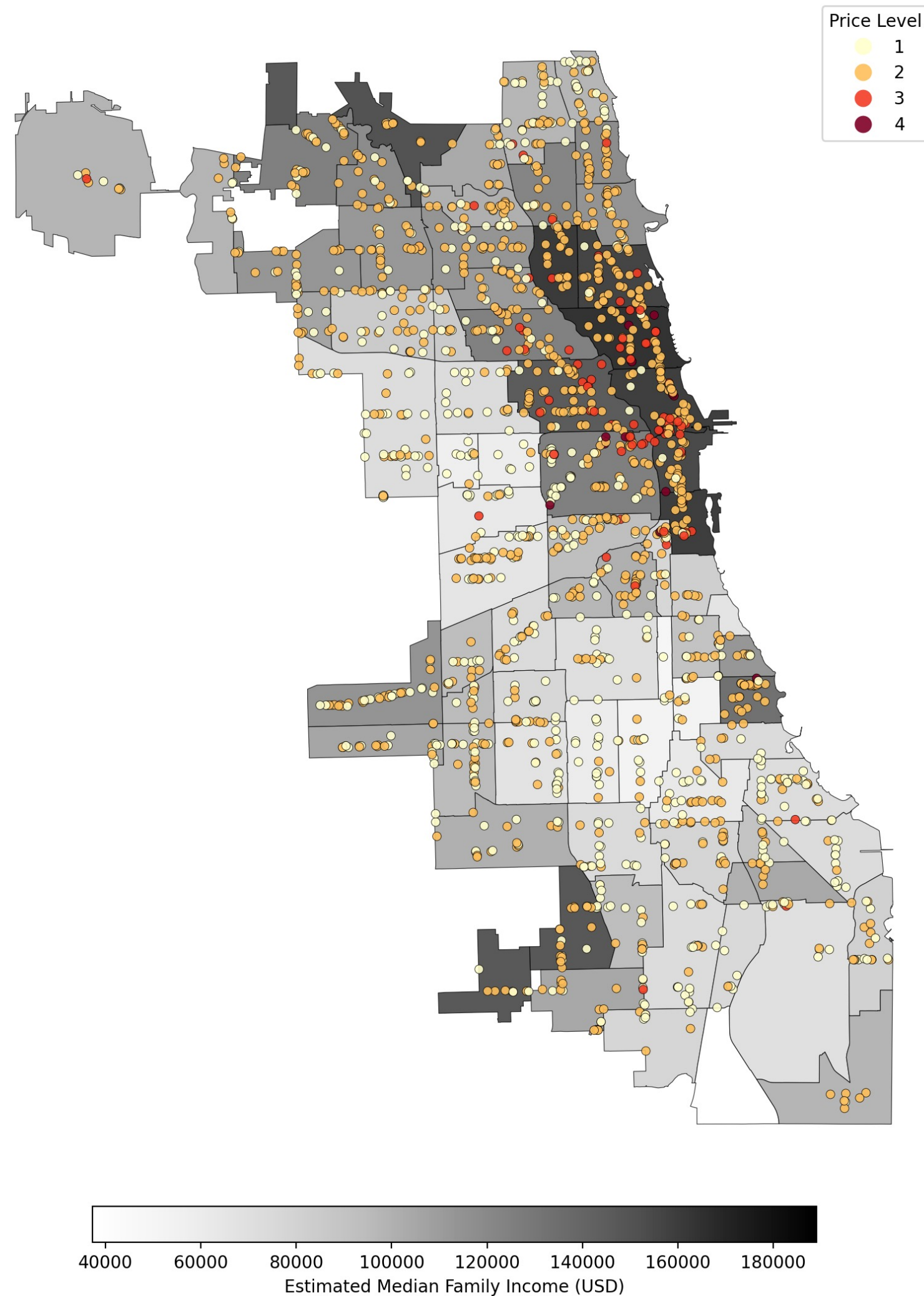
citywide grid sampling inside the Chicago boundary polygon



Plan sampling strategy before scraping

- Generate grid points
- Run nearby search around each grid point
- Deduplicate by restaurant id
- Adjust grid step size

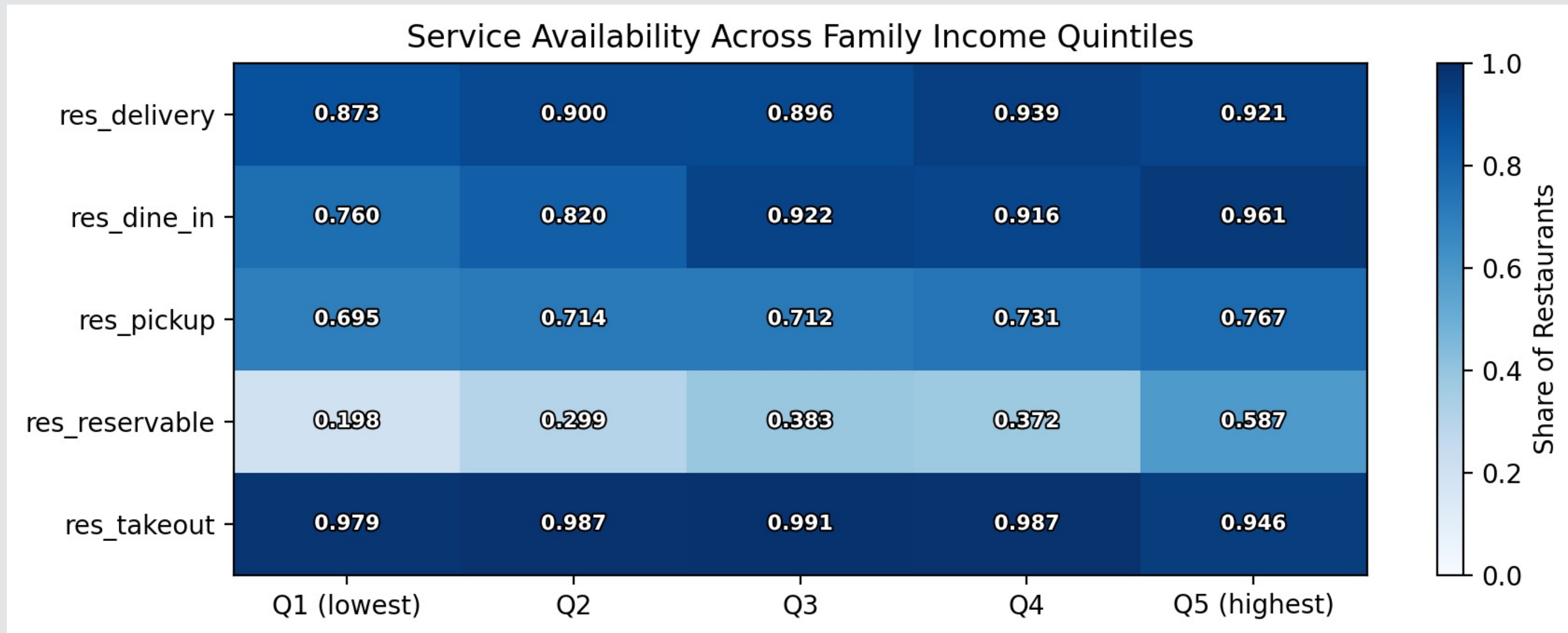
Spatial Correlation: Restaurant Price Levels & Community Income



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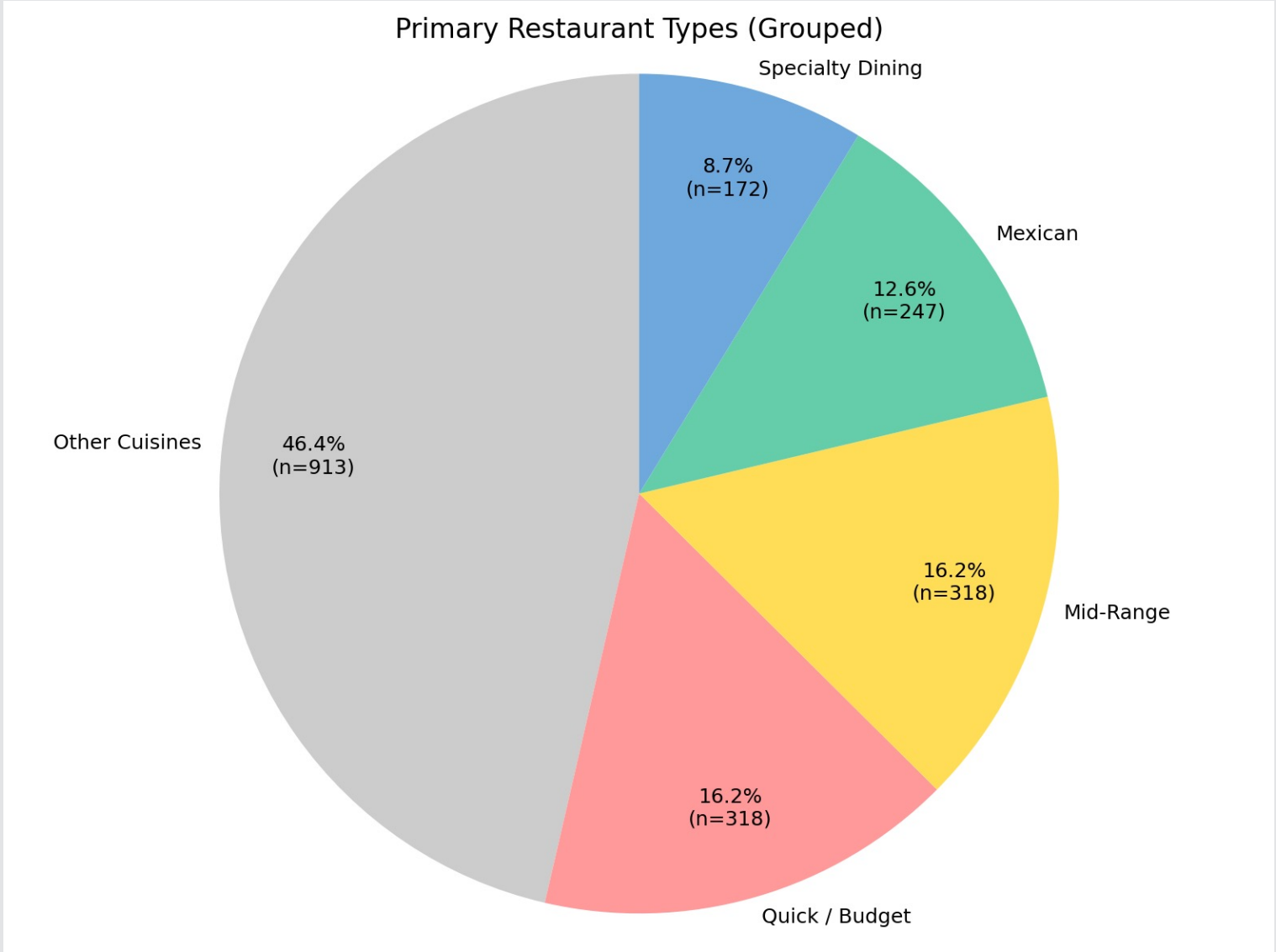
- background grey layer: income choropleth by community area
- point layer: restaurant points by price level (1–4)
- Higher-price restaurants cluster in higher-income neighborhoods
- Clear income gradient in restaurant price distribution

Service Availability Across Family Income Quintiles

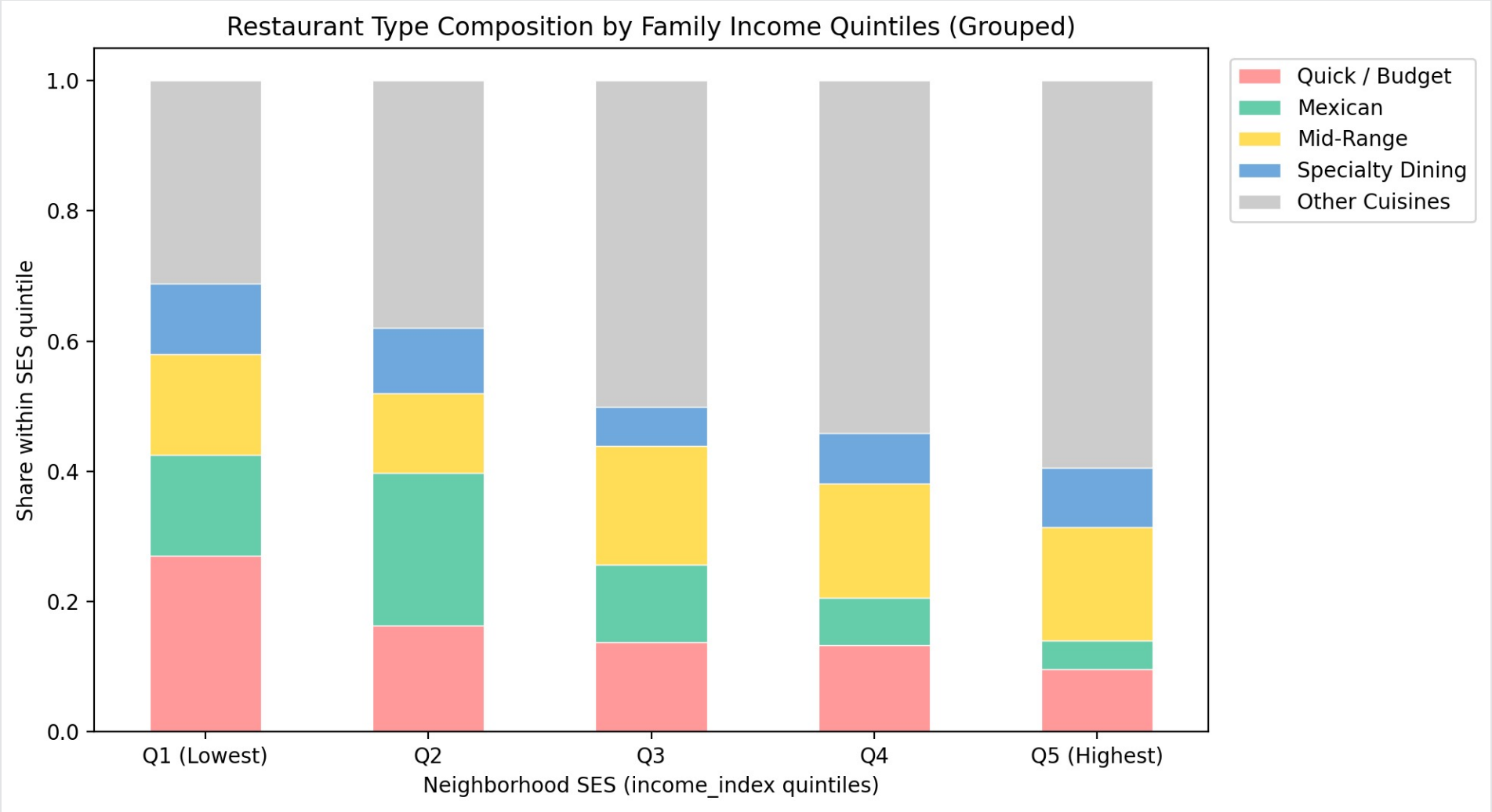


- Reservable shows the steepest income gradient (19.8% in Q1 → 58.7% in Q5)
- Dine-in availability also rises with income (76% → 96%)
- Takeout, delivery, and pickup are near-universal across all income levels

Primary Restaurant Types (Grouped)



Restaurant Type Composition by Family Income Quintiles (Grouped)



- Quick/Budget share declines with income (27% in Q1 → 12% in Q5)
- Mexican restaurants are more concentrated in lower-income areas
- Specialty Dining and Mid-Range grow as income rises

Predictors of Restaurant Price Levels

- Neighborhood income remains a significant predictor even after controlling for restaurant characteristics (Model 3)
- Reservable status is the strongest operational predictor of price level
- Takeout availability is negatively associated with price; delivery and dine-in show no effect
- Higher ratings and review counts are associated with higher prices

	Model 1	Model 2	Model 3
Income (z-score)	0.168*** (0.019)	0.123*** (0.019)	0.105*** (0.017)
Takeout	-0.721** (0.225)	-0.712*** (0.212)	-0.637** (0.201)
Delivery	-0.025 (0.055)	0.006 (0.058)	-0.000 (0.051)
Dine-in	-0.028 (0.043)	-0.067 (0.043)	0.042 (0.042)
Reservable	0.432*** (0.036)	0.386*** (0.031)	0.315*** (0.029)
Pickup	-0.000 (0.030)	-0.009 (0.031)	-0.009 (0.032)
Rating		0.202*** (0.033)	0.098*** (0.030)
Log reviews		0.056*** (0.014)	0.064*** (0.014)
Open hours (z)			-0.034* (0.015)
Late hours (z)			0.000 (0.017)
Type dummies	No	No	Yes
N	1803	1803	1800
R ²	0.296	0.323	0.391

Conclusions and Limitations

RQ1 (Price–SES):

- Higher-income community areas → higher average price levels
- This relationship remains even after controlling for restaurant type, service features, and rating-based “quality” signals

RQ2 (Characteristics–SES):

- High-priced restaurants and formal dining features (reservable & dine-in) are concentrated in higher-income central and lakefront areas of the city
- Lower-income areas have a higher share of low-tier options and a different cuisine mix

Limitation / next step:

- Future work could incorporate additional SES indicators (e.g., education levels or unemployment rate) to capture neighborhood socioeconomic conditions more comprehensively